

From State Recoverability to Event-Source Diagnosability: Physics-Based Inference From Sparse PMUs

JORGE LUIS MAYORGA TABORDA¹, YESSICA AFRICANO RODRIGUEZ¹, MASOUD BARATI²,
PAYMAN DEHGHANIAN³, AND DAVID FELIPE CELEITA RODRÍGUEZ⁴

¹Universidad de los Andes, Bogotá, Colombia

²Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA

³Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA

⁴Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

ABSTRACT Accurate recovery of an unmeasured voltage field does not imply that sparse phasor measurement units (PMUs) can identify the physical source of a disturbance. We formalize this distinction in a hybrid differential-algebraic model that separates hidden state, physical intervention, and measurement integrity. Functional observability characterizes the requested voltage field; a nuisance-profiled separation margin characterizes event-source diagnosability. Evidence comes from two frozen IEEE 39-bus campaigns using the same eight PMU buses but different simulators and contracts. On 100 PowerDynamics nonlinear test trajectories, a causal Kalman estimator reduces mean hidden-bus relative complex-voltage error from 0.3402% to 0.009634%; a preregistered functional residual correlates with bus-level error ($\rho = 0.5625$). A measurement-only regularized physical recentring closes 97.4%–98.3% of the nominal-center error gap on 60 operating-point-shifted plants despite nuisance-information rank 25/43. On a separate 690-trajectory ANDES event bank, a physics-guided hierarchy attains detection F1 0.989, event macro-F1 0.905, and physical-source Top-1 0.592 when sources are represented during training. Under complete source holdout, only 3 of 303 decisions are correct. The evidence establishes sparse-PMU state recoverability under the tested conditions and rejects source-generalization for the learned baseline; the proposed end-to-end joint estimator remains unvalidated.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet only a fraction of the buses and branches in a transmission network are normally instrumented [1], [2]. A disturbance may therefore originate outside the measured region, alter hidden voltages, and propagate to the available devices. At the same time, a missing frame or corrupted channel can resemble a physical transient. A useful diagnosis must answer three distinct questions: what is the network state, which physical or measurement event is present, and where did each applicable event originate?

Strong prior methods solve important subsets of this problem. Dynamic estimators combine physical models with synchrophasors [2], [3]; sparse-PMU classifiers identify events and faulted assets [4], [5]; and recent learning systems reconstruct hidden trajectories before localization or estimate state

and attack location from partial observations [6], [7]. These results make a broad “joint inference” novelty claim untenable. The unresolved contract considered here is narrower: causal inference in an under-observed transmission system when physical interventions and measurement-integrity failures can coexist, the source asset may be absent from source-labeled training, and the sensing geometry may make two candidate sources physically indistinguishable.

This distinction matters because event recognition, state reconstruction, and source localization have different information requirements. A generation change can be detected throughout the network while its source remains ambiguous. A dropped PMU identifies an integrity location directly but removes information needed to localize a simultaneous physical event. Likewise, a state estimate can have low point error although the nuisance intervention is not uniquely identifiable. Treating all three tasks as one multiclass label hides

these differences and encourages scores whose denominator does not match the claimed capability.

We represent a physical event as an intervention hypothesis $h = (c, \ell, \tau, \gamma)$ containing family, source, onset, and magnitude, and represent measurement integrity separately through a time-varying availability operator and an additive corruption process. Each physical candidate produces a counterfactual DAE response. Candidate comparison profiles the admissible initial-state and integrity nuisance directions. The resulting separation margin exposes source pairs that the available PMUs cannot distinguish and motivates a candidate set or abstention instead of a forced source.

The empirical record currently contains two complementary, frozen campaigns. The PowerDynamics campaign audits an IEEE 39-bus hybrid DAE, reconstructs 31 hidden complex bus voltages from eight voltage/current PMUs, and studies model mismatch and physical recentering. The separate ANDES campaign contains 690 trajectories spanning eight abnormal classes plus normal operation and evaluates a physics-guided hierarchical ExtraTrees baseline. The two campaigns share PMU buses and a network family, but not simulator semantics, state coordinates, or split contracts; their metrics are never pooled. The event study is presented as a baseline and failure analysis, not as evidence that the proposed joint posterior has already been executed.

The paper makes four contributions.

- 1) We formulate sparse-PMU monitoring as joint inference over hidden state, physical intervention, physical source, and measurement integrity. The eight event labels are mapped to composable physical and integrity variables rather than treated as eight unrelated phenomena.
- 2) We distinguish functional observability of the requested voltage field from event-source diagnosability. A finite-horizon affine-manifold distance gives a local theoretical limit: intersecting hypothesis manifolds admit nuisance configurations that no decision rule can distinguish uniformly.
- 3) We consolidate two audited evidence lines. On the state campaign, a preregistered structural residual predicts bus-level reconstruction difficulty and measurement-only regularized physical recentering recovers operating-point-shifted voltages. On the event campaign, multiscale and spatial features improve known-source localization, but complete source hold-out collapses to 3/303 exact decisions.
- 4) We define the reproducible comparison required for the next joint experiment: learned-only diagnosis, physics-only hypothesis inference, physics plus learned residual discrepancy, and diagnosability-aware inference with abstention, all evaluated on source-disjoint parents with explicit alarm exposure and PMU loss/join conditions.

The strongest result is also the main limitation. The state component is accurate under the tested nominal and operating-point-shift contracts, and the known-source event baseline attains detection F1 0.989 and event macro-F1 0.905.

Yet source localization does not transfer to assets excluded from training. The current manuscript therefore establishes the mathematical and empirical boundary that a joint physics-informed method must cross; it does not convert a proposed end-to-end estimator into a completed result.

The remainder of the paper reviews the closest estimation and localization contracts, defines the hybrid DAE and eight event labels, derives the observability and diagnosability limits, specifies the estimator hierarchy, documents both evidence campaigns, and reports the validated results and unresolved joint-inference gate.

II. RELATED WORK AND EXACT RESEARCH GAP

A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with high-rate synchronized measurements. The task-force review in [2] separates dynamic, tracking, forecasting-aided, and static formulations and identifies model uncertainty and measurement availability as persistent constraints. Robust Gaussian filters address outliers or non-Gaussian measurement noise [8], [9]; these are different failure modes from a stale network or equilibrium center. Nugroho *et al.* design robust observers for differential and algebraic DAE states, including unknown inputs and input-sensor failures [10]. Katanic *et al.* estimate the same state types in a partially known nonlinear DAE and derive a topological estimability condition [3]. Qu *et al.* derive an event-triggered joint state/unknown-input estimator for multi-machine grids [11], while Nguyen *et al.* use distributed state-input estimation in active networks [12]. These works preclude novelty based on “Kalman plus DAE” or joint state/input estimation. Their unknown inputs are not a multi-family physical-source bank with a simultaneous integrity state.

Learning methods recover states under sparse or heterogeneous measurements by encoding graph structure, power-flow relations, or a model-based estimator in the representation. Ngo *et al.* combine a graph neural network with physics-based state estimates [13]; Yue *et al.* couple graph learning to a robust distribution-system state estimator and include sparse-PMU line-outage identification [14]. Moshagh *et al.* address PMU-unobservable transmission state estimation with a topology-aware graph architecture and explicit tests under topology change, PMU failure, bad data, and non-Gaussian noise [15]. These are strong reconstruction alternatives, but a state estimator that tolerates an outage does not necessarily identify the outage source.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification has been addressed through subspace methods, engineered features, transfer learning, and graph models [24]–[27]. Cheng *et al.* demonstrate label-light online detection on a large field-PMU stream [28], while Abdel Hai *et al.* study transfer when only a small labeled set is available [29]. These works establish strong detection

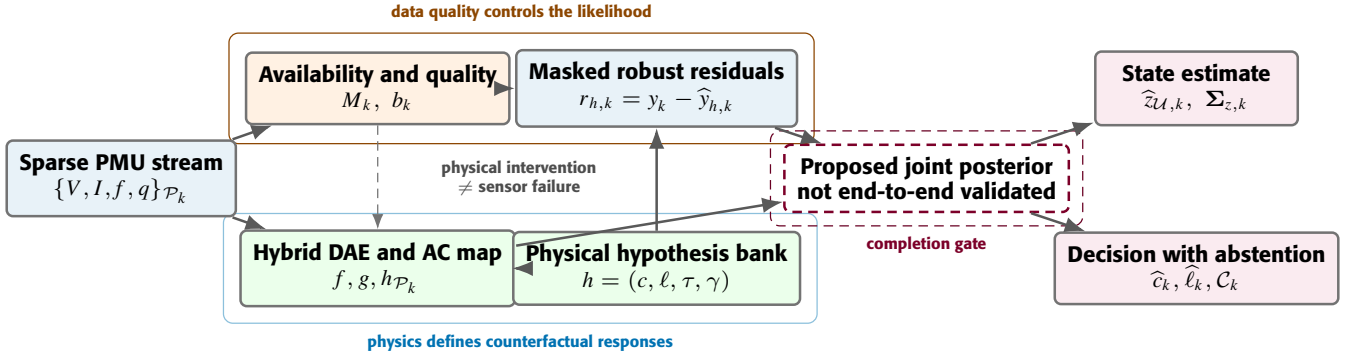


FIGURE 1. Joint inference contract. The physical model generates candidate state and PMU responses, while availability and quality govern which residuals are admissible. Physical interventions and measurement-integrity failures remain separate latent variables. The rightmost outputs distinguish hidden-state uncertainty from event/source ambiguity and permit a candidate set instead of a forced source. Solid arrows describe the proposed online information flow; this paper validates its state-estimation and historical diagnosis components separately.

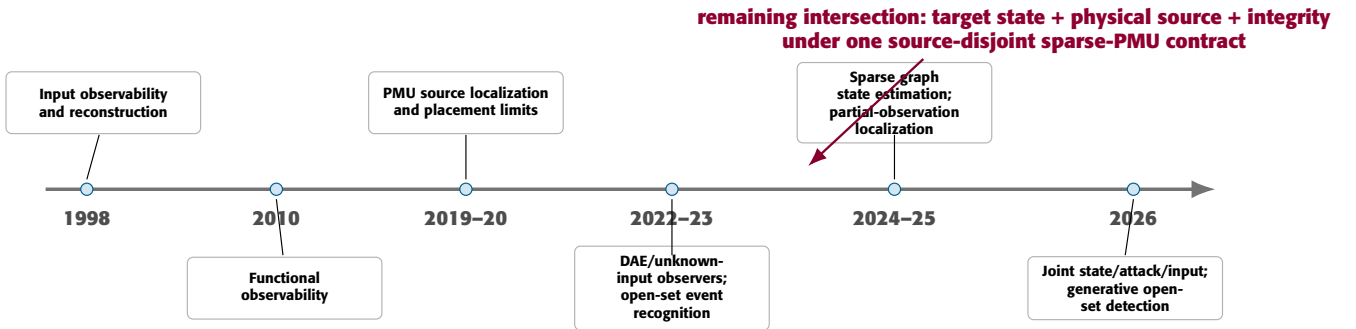


FIGURE 2. Methodological timeline and the unresolved intersection. Input/functional observability establish which latent quantities can be recovered [16], [17]; sparse-PMU localization and limits establish source-dependent sensing requirements [18], [19]; DAE observers and open-set classifiers address different parts of the problem [10], [20], [21]; recent graph, partial-observation, sparse-attack, and generative open-set methods extend those axes [7], [15], [22], [23]. The figure is a capability timeline, not a priority or performance ranking.

contracts, but event presence is not physical-source recovery. Pandey *et al.* jointly detect, classify, and localize events using synchrophasor measurements and system information [4]. Sparse-PMU methods locate faulted lines, generation losses, or outages using learned models and electrical structure [5], [18], [30], [31]. Group-sparse dictionaries use admittance-, impedance-, and injection-based response models to localize several disturbance types with limited measurements [32]; a spatiotemporal graph model can also embed event-type, topology, and localization-cost knowledge in its loss [33].

Open-set recognition supplies a different contract from source holdout. Kummerow *et al.* use Siamese recurrent embeddings to reject event types or locations absent from training [20]; their subsequent sigmoid-network formulation learns class-dependent rejection boundaries under limited and full PMU coverage [21]. Hu and Cheng use generative sliding-window evidence and decision fusion to assign unseen events to a dedicated open category [23]. These methods ask whether an observation belongs outside a learned class set. The present source-generalization question asks whether an untrained but physically enumerable bus or line can still be scored as a counterfactual source. Rejection and physical candidate ranking are complementary, not interchangeable.

Benchmark availability does not remove the contract prob-

lem. The pmuBAGE collection offers almost one thousand privacy-preserving generated event records learned from field data [34]; it is useful for standardized event analytics but does not by itself certify physical source holdout. Measurement integrity is also a distinct inference variable. Chatterjee *et al.* separate event-induced excursions from spurious outliers before robust PMU recovery [35], and Biswas *et al.* curate utility examples of missing, locked, quantized, and timing-affected synchrophasors [36]. Their evidence supports explicit integrity modeling instead of folding sensor failures into physical event classes.

These works show that limited PMU coverage is not by itself a new problem. The unresolved issue is the evaluation contract. Many classifiers learn a fixed list of asset labels, so testing another replica at a known source measures interpolation rather than source generalization. Kummerow and Bretschneider show that measurement noise, unseen operating points, and unseen disturbances materially change the evaluation of learned PMU classifiers [37]. Zhai *et al.* reconstruct full state and localize line attacks from partial observations using an LSTM across multiple benchmark networks [7]. Rimkus *et al.* reconstruct missing trajectories without fault-location labels before applying a localization method [6]. These are direct conceptual competitors and prevent us

from claiming to be the first to connect state reconstruction and localization. They do not jointly represent the eight physical/integrity event states used here or test the exact held-source, diagnosability, and model-mismatch contract.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [17], [38]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [16]; sparse strong observability extends the question to corrupted sensor sets [22], [39]. PMU placement based on empirical observability Gramians similarly treats sensor value as task dependent [40], [41]. For event sources, Jamei *et al.* derive placement-dependent performance limits for fault localization [19]; Conte *et al.* prove sparse-PMU fault-localization conditions that partition a radial grid into indistinguishable line clusters [42]; and Yildiz and Abur optimize sparse placement for event detection and identification [43]. These studies motivate the present separation between hidden-state reconstruction, input reconstruction, and pairwise source separation. They also impose a restraint: one prescribed eight-PMU set cannot support a general optimal-placement claim.

D. RESEARCH GAP AND CLAIM BOUNDARY

Table 1 exposes the remaining intersection rather than an absence of methods. The target is a source-agnostic physical hypothesis evaluated from sparse PMUs while (i) an unobserved state trajectory is estimated, (ii) physical interventions and measurement failures can coexist, (iii) source ambiguity is related to sensing geometry before a classifier is trusted, and (iv) generalization is tested on physical assets excluded from source-labeled training. A learned component is useful only if it improves model discrepancy or proposal efficiency without reverting to asset-label memorization.

The current evidence does not complete that intersection. It validates hidden-voltage estimation and physical recentering in PowerDynamics, and it audits a known-source event classifier plus source-transfer failure in ANDES. The present paper therefore centers the established boundary between recoverability and diagnosability. The proposed joint inference and separation-to-uncertainty experiment remain the next falsifiable step rather than a title-level performance claim.

III. JOINT STATE, EVENT, AND SOURCE PROBLEM

A. HYBRID DAE AND ELECTRICAL MEASUREMENT MODEL

Let $x(t) \in \mathbb{R}^{n_x}$ denote differential states, $z(t) \in \mathbb{R}^{n_z}$ algebraic states, $u(t)$ scheduled inputs, $q(t)$ network configuration, and $\theta_k = \theta_0 + G_\alpha \alpha_k$ physical parameters with a slowly varying operating coordinate α_k . A physical intervention $a_h(t)$ is indexed by hypothesis

$$h = (c, \ell, \tau, \gamma, d), \quad h \in \mathcal{H}, \quad (1)$$

where c is event family, ℓ is a bus or branch source, τ is onset, γ is severity, and d describes the allowed temporal profile. The plant is

$$\dot{x}(t) = f(x, z, u, q, a_h; \theta_k) + w_x(t), \quad (2)$$

$$\mathbf{0} = g(x, z, u, q, a_h; \theta_k), \quad (3)$$

$$y_k = M_k[h_{\mathcal{P}_k}(x_k, z_k, q_k; \theta_k) + b_k] + v_k. \quad (4)$$

Here M_k selects the channels that are available at time k , b_k is a measurement-corruption process, and v_k is measurement noise. A dropout changes M_k ; bad data change b_k ; neither is a physical input to (2)–(3). This separation permits a physical event and a measurement failure to coexist.

The latent variables have deliberately different time scales and meanings. The pair (x, z) is the fast dynamic–algebraic state; α is slow operating-point or parameter drift; a_h is an abrupt, sparse physical intervention; q is a discrete network mode; and (M, b) describes data availability and corruption. A colored model-discrepancy state c_k may enter either the process or measurement residual, as tested in Section V, but is not interchangeable with an event.

Within a smooth mode, assume the DAE is locally index one, so g_z is nonsingular after the reference-angle constraint is imposed. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1} \left(g_x f + g_u \dot{u} + g_\theta \dot{\theta} \right). \quad (5)$$

Equation (5) is not applied across a fault insertion or topology switch, where algebraic variables can jump and the model must be segmented. Differential states remain continuous unless a simulator-specific reset map is declared.

The algebraic network core is

$$I = Y_{\text{bus}}(q, \theta)V, \quad S_i = P_i + jQ_i = V_i I_i^*. \quad (6)$$

Writing $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$ gives

$$P_i = |V_i| \sum_j |V_j| [G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}], \quad (7)$$

$$Q_i = |V_i| \sum_j |V_j| [G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}], \quad (8)$$

where $\vartheta_{ij} = \vartheta_i - \vartheta_j$. A line outage changes the branch primitives and therefore Y_{bus} ; a bus fault adds a shunt-admittance intervention; generation and load changes modify the appropriate active/reactive injection equations. For a PMU at one terminal of branch (i, j) ,

$$I_{ij} = Y_{ff,ij} V_i + Y_{ft,ij} V_j, \quad I_{ji} = Y_{tf,ij} V_i + Y_{tt,ij} V_j. \quad (9)$$

The orientation, transformer tap, line charging, and shunt convention in (9) are part of the inference contract; replacing terminal current with bus-injection current changes the likelihood.

At the acquisition boundary, native measurements can include V_{abc} , I_{abc} , frequency, ROCOF, quality flags, and availability. Positive-sequence phasors define the balanced network likelihood used here. Negative- and zero-sequence components, when available, remain unbalance or integrity

TABLE 1. Capability comparison for the closest methodological families. “Held source” means that the physical asset is absent from source-labeled training or tuning. Entries describe published contracts, not a cross-paper performance ranking.

Work	State target	Event/source target	Physics mechanism	Sparse/partial sensing	Held source	Integrity separate
Katanic <i>et al.</i> [3]	Differential and algebraic DAE states	Unknown injectors, not event taxonomy	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho <i>et al.</i> [10]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Robust DAE observer	PMU voltage/current	N/A	Failure robustness, not compositional source
Qu <i>et al.</i> [11]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo <i>et al.</i> [13]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [14]	Distribution-system state	Line outage	Grid-model-informed graph learning and robust SE	Yes	Not stated	No
Moshtagh <i>et al.</i> [15]	Transmission-system state	Topology variation, not source output	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad-data tests
Pandey <i>et al.</i> [4]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [5]	None	Fault detection and location	OLS/CNN with strategic PMU placement	Yes	Not stated	No
Li <i>et al.</i> [33]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow <i>et al.</i> [21]	None	Known/unknown event type and location	Siamese distance with learned rejection	Limited/full PMU sets	Open-set rejection	Rejection boundary
Hu and Cheng [23]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No physical-source posterior
Rimkus <i>et al.</i> [6]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	No source labels for imputer	Corruption via imputation
Zhai <i>et al.</i> [7]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmark contracts	Attack target only
This paper	Hidden voltage field; state uncertainty	Eight event labels; physical and integrity sources	DAE hypotheses, profiled separation, residual learning proposed	Eight prescribed PMU buses	Yes; learned baseline fails	Yes

evidence rather than being mixed into the positive-sequence state. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2\angle V_i}{dt^2}, \quad (10)$$

but an implemented likelihood must use the causal estimator and filtering semantics of the corresponding PMU backend.

B. SPARSE PMUS AND REQUESTED STATE FUNCTION

Both evidence campaigns use PMU buses

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad |\mathcal{P}| = 8. \quad (11)$$

The unobserved set is $\mathcal{U} = \{1, \dots, 39\} \setminus \mathcal{P}$, with $|\mathcal{U}| = 31$. The state-estimation campaign stacks rectangular voltage and controlled terminal-current components,

$$y_{\mathcal{P},k} = \text{col}_{p \in \mathcal{P}} [\Re V_{p,k}, \Im V_{p,k}, \Re I_{pl,k}, \Im I_{pl,k}], \quad (12)$$

and requests the hidden complex-voltage function

$$z_{\mathcal{U},k} = \text{col}_{i \in \mathcal{U}} [\Re V_{i,k}, \Im V_{i,k}] \in \mathbb{R}^{62}. \quad (13)$$

The event campaign also uses voltage angle, frequency, availability, and, in its longer-window implementation, current and ROCOF. Measurement semantics are kept campaign-specific rather than declaring channels equivalent across simulators.

C. EIGHT EVENT LABELS AS COMPOSABLE STATES

Table 2 defines the eight abnormal labels in the historical event bank. The scientific variables are not the label numbers. Classes 1–4 are single physical interventions; classes 5 and 7 are measurement-integrity states; classes 6 and 8 are compositions. Class 8 is a prescribed mixed event in the available simulations, not evidence of open-set recognition.

Table 3 states how the four physical families enter the candidate DAE. It is an operator contract, not a claim that the proposed posterior has already solved every family.

The internal decision at time k is

$$\left(\hat{z}_{\mathcal{U},k}, \hat{\Sigma}_{z,k}, \hat{c}_k, \hat{\ell}_k^{\text{phys}}, \hat{m}_k, \hat{\ell}_k^{\text{meas}}, c_k \right), \quad (14)$$

where m_k denotes the integrity state and $C_k \subseteq \mathcal{H}$ is a credible candidate set. A location is empty when it does not apply. The candidate set is required because limited observability can make a forced Top-1 decision scientifically unjustified.

D. CORRECTNESS AND EVALUATION UNITS

State accuracy uses the mean hidden-voltage relative complex error (HVRE)

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (15)$$

This trajectory-and-bus aggregate is reported as a percentage. It is TVE-style but is not the standard per-frame synchrophasor TVE conformity statistic. Event detection, event-family macro-F1, exact physical Top-1/Top-3, and integrity-source Top-1 use their own declared populations. The joint correctness indicator is one only when every applicable component of (14) is correct; a correct event family cannot conceal a wrong source. Confidence intervals use an independently generated trajectory, event parent, or rebuilt plant as the statistical unit, never individual 30-Hz samples.

The causal contract permits only $y_{0,k}$ at time k . A fixed-lag smoother, complete-window statistic, or centered temporal feature is retrospective and is labeled accordingly. Source generalization requires that physical source ℓ be absent from all source-labeled training and tuning, not merely that the test trajectory use a new random seed.

IV. RECOVERABILITY AND DIAGNOSABILITY UNDER SPARSE PMUS

A. FUNCTIONAL OBSERVABILITY OF THE REQUESTED VOLTAGE FIELD

Linearizing the index-reduced DAE around a verified equilibrium and sampling at $\Delta t = 1/30$ s gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \quad (16)$$

$$\delta y_k = C \delta x_k + v_k, \quad (17)$$

$$\delta z_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k. \quad (18)$$

The reduction assumes a regular local algebraic Jacobian after the reference mode is fixed. Here $\delta x_k \in \mathbb{R}^{114}$ contains reduced differential coordinates, C maps them to 32 rectangular voltage/current PMU channels, and $L_{\mathcal{U}}$ maps them to 62

rectangular hidden-voltage components. These matrices are local and hypothesis dependent after a topology change.

For a horizon of N transitions, define

$$\mathcal{O}_N = \text{col} (C, CA_d, \dots, CA_d^N). \quad (19)$$

Full-state observability requires $\text{rank}(\mathcal{O}_N) = 114$. The requested voltage at time $r \leq N$ instead has map $L_r = L_{\mathcal{U}} A_d^r$. Recovering the complete local target sequence uses $\mathcal{L}_N = \text{col}(L_0, \dots, L_N)$.

Proposition 1 (Finite-horizon functional reconstruction): For the noise-free local system, the target at time r is uniquely determined by the output sequence through N if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \quad (20)$$

The complete target sequence is uniquely determined if and only if $\ker \mathcal{O}_N \subseteq \ker \mathcal{L}_N$.

Proof: Two initial states produce the same output history exactly when their difference belongs to $\ker \mathcal{O}_N$. Their requested outputs at time r agree for every such pair exactly when the difference also belongs to $\ker L_r$. Stacking the maps gives the sequence statement. Equivalently, the rows of the requested map lie in the row space of $\mathcal{O}_N$. ■

This standard functional-observability result [17], [38] fixes the target and time index before any estimator is assessed. It also shows why a low full-state rank alone does not preclude recovery of the hidden-voltage function.

B. NUMERICAL AUDIT FOR THE PRESCRIBED EIGHT-PMU CASE

For the anchor-time target used by the frozen structural audit, define

$$\Pi_N = \mathcal{O}_N^\dagger \mathcal{O}_N, \quad (21)$$

$$r_F(N) = \|L_{\mathcal{U}}(I - \Pi_N)\|_F, \quad (22)$$

$$r_{F,i}(N) = \|L_i(I - \Pi_N)\|_F. \quad (23)$$

A zero residual is an exact numerical condition only when the matrix scaling and rank decision are defensible. A nonzero value certifies that the computed observable subspace does not contain every requested anchor-time direction.

The frozen implementation formed $\mathcal{O}_N^\dagger \mathcal{O}_N$ and retained eigenvalues above 10^{-9} of its maximum. Figure 3 exposes, rather than hides, the sensitivity of that choice. The pseudo-rank reaches 54 at $N = 30$ but falls to 39 at $N = 180$, although the exact rank of a vertically augmented observability matrix cannot decrease. This behavior results from Gramian squaring, unstable-mode scaling, and a relative threshold; it prevents the pseudo-rank from being treated as an algebraic certificate. The target residual is also nonzero at every tested horizon. By contrast, the reported noise-amplification diagnostic decreases as more temporal information is accumulated.

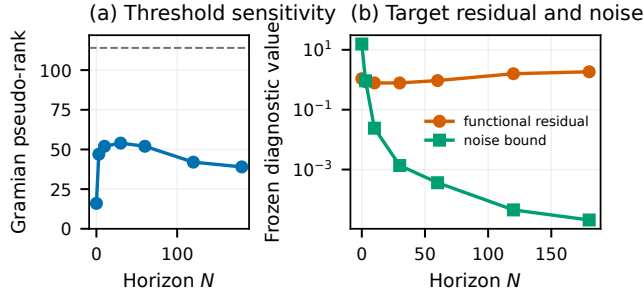
Horizons $N \in \{0, 3, 10, 30, 60, 120, 180\}$ correspond to 0–6 s. The nonlinear nominal trajectories contain 91 frames, or 3 s, so $N = 180$ is not a matched causal horizon for those records. Its bus ranking was nevertheless exported before

TABLE 2. Event ontology and historical ANDES inventory. Each source configuration has five replicas. “Physical source” and “integrity source” are separate outputs.

ID	Operational meaning	Physical intervention c	Physical source ℓ	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
Total					690

TABLE 3. Physical intervention operators in the candidate bank. Algebraic jumps are resolved by the post-event network constraints; differential-state resets are absent unless explicitly stated.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive, $Y_{bus}^+ = Y_{bus}^- + y_{fe} e_{\ell} e_{\ell}^T$, for the declared clearing interval	admissible buses	complex fault admittance or resistance/reactance	x continuous; z may jump
Line outage	Set branch status $q_{ij} : 1 \mapsto 0$ and remove its tap, series, charging, and shunt primitives from Y_{bus}	in-service branches	discrete status and switching time	x continuous; z may jump
Generation change	Modify scheduled active/reactive injection or a declared governor reference at generator bus ℓ	generator buses	$\Delta P_G, \Delta Q_G$, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model at load bus ℓ	load buses	$\Delta P_L, \Delta Q_L$, ZIP convention	continuous input-driven response

**FIGURE 3.** Frozen local observability audit for PMU buses {2, 5, 6, 10, 19, 22, 29, 39}. (a) Gramian pseudo-rank under the stated relative threshold; its nonmonotonicity is numerical and rules out an exact-rank claim. (b) Nonzero anchor-time functional residual and the trace-style noise-amplification bound. These quantities diagnose the local model; they do not prove nonlinear recoverability.

nonlinear scoring and is retained only as a preregistered, cross-horizon structural diagnostic. The empirical test correlates $r_{F,i}(180)$ with bus-level HVRE; it does not convert the six-second local audit into a three-second observability proof.

C. EVENT-SOURCE DIAGNOSABILITY AS AFFINE-MANIFOLD SEPARATION

State recoverability and source diagnosability have different nuisance spaces. Under candidate h , stack the available channels over H samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H} \xi_h + \mathcal{E}_{S,H} \beta_h + \varepsilon_H, \quad \varepsilon_H \sim \mathcal{N}(\mathbf{0}, R_H), \quad (24)$$

where $s_{h,H}$ is the intervention response, ξ_h is the locally admissible initial-state deviation, and $\mathcal{E}_{S,H} \beta_h$ is corruption on a fixed support S . Allowing each candidate to refit nuisance variables produces the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{col}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (25)$$

This candidate-specific form is required for topology-changing events because $\mathcal{O}_{i,H}$ and $\mathcal{O}_{j,H}$ need not agree.

Let $\Delta s_{ij} = s_{i,H} - s_{j,H}$ and

$$\mathcal{N}_{ij,S} = [\mathcal{O}_{i,H}, -\mathcal{O}_{j,H}, \mathcal{E}_{S,H}]. \quad (26)$$

The R_H^{-1} -orthogonal nuisance projector and the pairwise separation margin are

$$P_{ij,S}^\perp = I - \mathcal{N}_{ij,S} (\mathcal{N}_{ij,S}^\top R_H^{-1} \mathcal{N}_{ij,S})^\dagger \mathcal{N}_{ij,S}^\top R_H^{-1}, \quad (27)$$

$$D_{ij,S}^{(P)}(H) = \frac{1}{2} \|P_{ij,S}^\perp \Delta s_{ij}\|_{R_H^{-1}}^2. \quad (28)$$

Equation (28) is half the minimum squared weighted distance between the two affine manifolds. It becomes the Gaussian mean-separation term only after nuisance values are fixed at the closest points. If all candidates are conditioned on one common pre-event state, the corresponding contract replaces the two independent state blocks in (26) by $\mathcal{O}_{i,H} - \mathcal{O}_{j,H}$. The paper does not mix these two contracts.

Proposition 2 (Local composite-hypothesis indistinguishability): If $D_{ij,S}^{(P)}(H) = 0$, then $\mathcal{M}_i(S)$ and $\mathcal{M}_j(S)$ intersect. Hence there exist admissible nuisance values under candidates i and j that induce the same Gaussian measurement law. No decision rule can distinguish the pair uniformly over the

declared nuisance set; with equal prior mass on those two parameter points, its average error is at least $1/2$.

Proof: The margin is zero exactly when $\Delta s_{ij} \in \text{col}(\mathcal{N}_{ij,S})$, which is equivalent to an intersection of the two affine mean sets. At the intersecting nuisance values, the means and covariance are identical. The observation law is therefore identical under both parameter points, so the two conditional error probabilities sum to one. ■

When the corruption support is unknown but at most q channels are affected, the conservative margin is

$$D_{ij}^{(q)}(H) = \min_{S: |S| \leq q} D_{ij,S}^{(\mathcal{P})}(H). \quad (29)$$

This union-of-subspaces formulation connects source localization to sparse strong observability and unknown-input reconstruction [16], [22], [39]. It also states what an integrity-robust diagnosability claim must survive.

Corollary 1 (Monotonicity under an added PMU): For the same pair, covariance model, nuisance contract, and integrity budget, adding a conditionally independent PMU block cannot reduce the profiled margin:

$$D_{ij,S}^{(\mathcal{P} \cup \{p\})}(H) \geq D_{ij,S}^{(\mathcal{P})}(H). \quad (30)$$

For every nuisance vector, the augmented squared residual is the old nonnegative residual plus the new PMU residual. Taking the minimum preserves the inequality. Equality with a sum of separately profiled margins does not hold in general because the same nuisance variables must explain both sensor blocks; additivity requires fixed simple hypotheses or additional separability conditions.

For fixed closest nuisance values and unit-severity signature, the equal-prior Gaussian error is

$$P_{e,ij}^* = \Phi\left(-\sqrt{D_{ij}/2}\right), \quad |\gamma|_{\min} = \frac{\sqrt{2} \Phi^{-1}(1-\beta)}{\sqrt{D_{ij}(1)}} \quad (31)$$

to target $P_e^* \leq \beta$ when the signature scales linearly with severity γ . This local severity limit is not applied across topology switches or nonlinear fault saturation. For simultaneous events with projected unit-signature matrix $\tilde{S}_A$, their amplitudes are locally identifiable only if $\text{rank}(\tilde{S}_A) = |A|$ after the same nuisance projection.

For a calibrated source posterior, define the smallest candidate set satisfying

$$C_k(\alpha) = \arg \min_{C \subseteq \mathcal{H}} \left\{ |C| : \sum_{h \in C} p(h | y_{0:k}) \geq 1 - \alpha \right\}. \quad (32)$$

A diagnosability-aware system abstains from exact Top-1 when $|C_k| > 1$ or when its closest competing margin falls below a development-set threshold. The pending common-contract experiment must test whether the margin predicts candidate-set size, calibrated uncertainty, and source-resolution delay under PMU loss and join. The current event bank has not supplied that evidence.

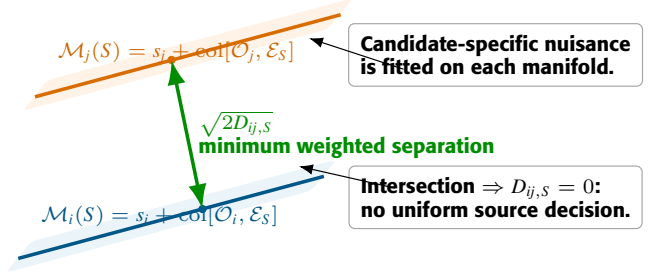


FIGURE 4. Local source-diagnosability geometry. Each candidate forms an affine set after state and integrity nuisance are profiled. The shortest weighted separation, not the raw signature difference, determines the local margin. Intersecting sets admit indistinguishable candidate–nuisance pairs.

V. PHYSICS-BASED STATE ESTIMATION AND JOINT INFERENCE

A. EXACT TARGET AND A GENERATIVE CANDIDATE MODEL

Let $\zeta_k = L(x_k, z_k)$ be the electrical quantity requested by the application. Under squared loss, the exact Bayesian target is

$$\hat{\zeta}_k^{\text{MMSE}} = \mathbb{E}[\zeta_k | y_{0:k}], \quad \hat{h}_k^{\text{MAP}} = \arg \max_{h \in \mathcal{H}} p(h | y_{0:k}). \quad (33)$$

This statement is an optimality definition, not an implemented-algorithm claim.

For candidate $h = (c, \ell, \tau, \gamma, d)$, assume $x_0 \sim \mathcal{N}(\bar{x}_0, P_0)$, Gaussian process and measurement noise, the hard constraint $g_h(x_t, z_t) = \mathbf{0}$, and a Laplace prior on active corruption coefficients. With

$$r_{h,t} = M_t[y_t - h p_t(x_t, z_t) - b_t], \quad (34)$$

the negative twice-log joint density conditional on h , up to constants and the declared corruption-support prior, is

$$\begin{aligned} \mathcal{J}_h^\circ &= \|x_0 - \bar{x}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^k \|r_{h,t}\|_{R_t^{-1}}^2 \\ &+ \sum_{t=0}^{k-1} \|x_{t+1} - F_h(x_t, z_t)\|_{Q^{-1}}^2 + 2\lambda_b \sum_{t=0}^k \|b_t\|_1, \end{aligned} \quad (35)$$

subject to the DAE constraints. Crucially, $\mathcal{J}_h^\circ$ does not contain the hypothesis prior. The physical family and source enter through F_h, g_h , and the network/measurement operators rather than as an asset-label head.

The evidence is the constrained integral

$$p(y_{0:k} | h) = \int_{g_h=0} \exp[-\mathcal{J}_h^\circ/2] dX dZ dB. \quad (36)$$

A Laplace approximation on locally identifiable directions gives

$$p(h | y_{0:k}) \propto p(h) \det^+(H_h)^{-1/2} \exp[-\mathcal{J}_h^\circ(\hat{X}_h, \hat{Z}_h, \hat{B}_h)/2], \quad (37)$$

so the prior appears exactly once. The pseudo-determinant requires the active constraint and corruption support to be fixed locally. If a Huber loss, a soft DAE penalty, or an

omitted determinant is used without a corresponding normalized density, the result is a penalized evidence score, not a posterior probability.

The learned component is deliberately restricted. The earlier engineered feature system can propose a candidate set or learn a correction $\psi_\phi(r_{h,0:k})$ to physical log evidence. It must consume candidate residuals, not source identity alone, and must be tested on held-out physical sources. The required ablation is learned-only, physics-only, physics plus learned discrepancy, and the complete diagnosability-aware rule.

B. NOMINAL ESTIMATOR HIERARCHY

The estimators form an evidence ladder rather than a collection of unrelated algorithms. B0 retains only the equilibrium, B1 adds instantaneous PMU conditioning, B2 adds causal state propagation, and B3 tests retrospective temporal information. When the equilibrium itself becomes stale, the separate E06-H level estimates a physically admissible operating point through regularized nonlinear recentering before any sequential extension is claimed. A full nonlinear Hybrid-DAE fixed-lag posterior is therefore an enclosing research formulation, not a result of this paper.

All nominal estimators use the same equilibrium y_0 , hidden-voltage center $z_{\mathcal{U},0}$, sampled dynamics A_d , PMU map C , and hidden-output map $L_{\mathcal{U}}$. This shared model isolates the value of measurement updating and temporal propagation.

The equilibrium baseline B0 returns

$$\hat{z}_{\mathcal{U},k}^{\text{B0}} = z_{\mathcal{U},0}. \quad (38)$$

It measures how much of the score can be obtained by remaining at the nominal power-flow solution.

The snapshot estimator B1 computes a prior-conditioned linear minimum-mean-square-error (LMMSE) update independently at every frame,

$$K_{\text{LMMSE}} = P_0 C^\top (C P_0 C^\top + R)^{-1}, \quad (39)$$

$$\hat{\delta x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0), \quad \hat{z}_{\mathcal{U},k}^{\text{B1}} = z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{\delta x}_k^{\text{B1}}. \quad (40)$$

B1 uses network-constrained cross-covariance but no state carried from frame $k - 1$.

Conditional on one linear-Gaussian candidate with no corruption, B2 is not an ad hoc module: the Kalman recursion computes the exact conditional mean and covariance in (33). Candidate uncertainty, nonlinear switching, unknown integrity support, and non-Gaussian discrepancy are what make the enclosing hybrid problem approximate.

The causal Kalman estimator B2 applies the classical linear-Gaussian recursion [44]:

$$\hat{x}_{k|k-1} = A_d \hat{x}_{k-1|k-1}, \quad (41)$$

$$P_{k|k-1} = A_d P_{k-1|k-1} A_d^\top + Q, \quad (42)$$

$$\nu_k = (y_k - y_0) - C \hat{x}_{k|k-1}, \quad (43)$$

$$S_k = C P_{k|k-1} C^\top + R, \quad (44)$$

$$K_k = P_{k|k-1} C^\top S_k^{-1}, \quad (45)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \nu_k. \quad (46)$$

The covariance update uses the Joseph form. The hidden estimate is $z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{x}_{k|k}$. B2 is causal because only $y_{0:k}$ enters the result at k .

B3 is an exact fixed-lag Gaussian smoother, following the linear Gaussian backward-recursion principle in [45], evaluated at lags $\ell \in \{0, 1, 3, 5, 10, 15, 30, 60\}$ frames. It uses measurements through $k + \ell$ to refine the estimate at k and is therefore retrospective by $\ell/30$ seconds. Dense and cached implementations agree numerically on representative lags. B3 tests whether temporal information beyond the causal update improves the hidden-voltage target; it is not used to support a real-time claim.

C. MARGINAL CALIBRATION AND COLORED DISCREPANCY

For the hidden two-component voltage error at bus i , B2 provides

$$\Sigma_{i,k} = L_i P_{k|k} L_i^\top. \quad (47)$$

The original covariance is U_0 . A two-block temperature model U_2 is fitted on an independent calibration split:

$$\Sigma_{i,k}^{U_2} = T \Sigma_{i,k} T^\top, \quad T = \text{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}). \quad (48)$$

The temperatures minimize Gaussian negative log likelihood (NLL) on calibration data and are frozen before TEST. Coverage, NLL, normalized innovation squared (NIS), and temporal innovation correlation are reported separately; nominal marginal coverage alone is not called complete calibration.

The residual study considers a latent Gauss–Markov process

$$c_{k+1} = F_c c_k + \eta_k. \quad (49)$$

In D2-M, c_k enters the PMU measurement residual through a calibration-fitted low-rank basis. In D2-P, it enters a sparse physical process dictionary. Basis, rank, F_c , and latent covariance are selected using calibration records only. The experiment asks whether the observed low-rank structure improves reconstruction, hidden uncertainty, or innovation whiteness. It does not assume that a lower innovation autocorrelation is sufficient.

D. MODEL-ADEQUACY STATISTIC AND RECENTRING ABLATIONS

Let μ_ν and Σ_ν be the mean and shrunk covariance of B2 innovations on the nominal calibration set. The per-frame adequacy statistic is the five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^\top \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (50)$$

Mismatch labels and hidden truth do not enter (50); hidden truth is used only after inference to label frames whose hidden HVRE exceeds the frozen evaluation threshold. AUROC and AUPRC therefore measure whether observed innovations anticipate unacceptable hidden error, not whether the detector identifies a mismatch family.

Three variants diagnose the failure mechanism without merging their evidence contracts. R1 replaces the nominal

equilibrium with the true plant equilibrium and is explicitly oracle. R2-A estimates an eight-dimensional grouped observed-bus voltage offset using a 60-frame trailing window, nominal matrices, ridge regularization, and only PMU measurements. It is a causal negative ablation: its center/state cross-covariance is neglected and it does not enforce the nonlinear AC equations.

The corrected regularized physical estimator addresses M6 operating-point mismatch at one equilibrium snapshot. Let

$$d = \text{col}(\Delta P_{PQ}, \Delta Q_{PQ}, \Delta P_{PV}) \in \mathbb{R}^{43} \quad (51)$$

collect active and reactive changes at 17 admissible PQ loads and active-power changes at nine PV buses. Voltage is parameterized as $V(\theta, \rho) = \exp(\rho + j\theta)$. Defining the scaled PMU residual as $e_y(V) = D_y^{-1}[h_P(V) - y_P]$, the estimator solves

$$(\hat{V}, \hat{d}) = \arg \min_{V, d} \mathcal{J}(V, d), \quad (52)$$

$$\mathcal{J}(V, d) = \|e_y(V)\|_2^2 + \gamma^2 \|g_{AC}(V, d)\|_2^2 + \frac{\lambda}{q_d^2} \|d\|_2^2.$$

where h_P is the audited voltage/terminal-current operator, g_{AC} includes PQ balance, explicit PV-magnitude constraints, solved PV reactive power, and bus 31 as Slack, and D_y scales voltage and current components separately. The weights are $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$, selected on a disjoint DEV split; continuation initialization is then frozen for TEST. Neither the true nuisance vector nor the hidden voltages enter the TEST solve.

Because d may be non-unique, evaluation distinguishes nuisance information from target information. If J_{yd} and J_{zd} are the observed-PMU and hidden-voltage Jacobians with respect to d after eliminating voltage through the AC constraints, the local target residual is

$$r_{d \rightarrow z} = \frac{\|J_{zd} N(J_{yd})\|_F}{\max(\|J_{zd}\|_F, \epsilon)}, \quad (53)$$

where $N(J_{yd})$ spans the PMU-uninformed nuisance nullspace. Exact target identifiability would require a zero residual. The experiment instead tests the weaker empirical question: whether (52) reconstructs the hidden voltage accurately despite rank-deficient nuisance information, and whether its Laplace covariance is credible.

VI. VALIDATED BENCHMARK AND EXPERIMENTAL PROTOCOL

A. BENCHMARK PROVENANCE AND NUMERICAL GATES

The primary plant uses Julia 1.11.9, PowerDynamics.jl 5.0.0, and NetworkDynamics.jl 1.3.0 [46]. It contains 39 buses, 46 branches, 19 loads, 10 sixth-order Sauer–Pai machines, nine AVR Type-I controllers, and nine TGOV1 governors. The descriptor realization has 192 coordinates: 114 differential and 78 algebraic. These parameters are package-defined tutorial data. We therefore classify the plant as a numerically validated pedagogical benchmark rather than an authoritative representation of an external grid.

The validation ladder prevents estimator performance from hiding an upstream modeling error. A same-plant static translation was audited after correcting tap orientation, shunts, line charging, and full- Y_{bus} injection accounting. Final maximum discrepancies were 1.14×10^{-13} pu for a Y_{bus} element, 1.73×10^{-7} pu for voltage magnitude, 1.40×10^{-5} degrees for angle, and 1.71×10^{-3} MVA for terminal flow. The controlled PMU operator then achieved maximum complex-voltage and analytic/backend terminal-current discrepancies of 2.71×10^{-7} and 1.43×10^{-14} pu, respectively.

Dynamic validation checks the equilibrium, local modes, nonlinear trajectories, and linearization. The DAE equilibrium residual is 4.66×10^{-13} . After removing the reference mode, the least-damped physical pair is $-0.09818 \pm j0.42948 \text{ s}^{-1}$. Nonlinear Rodas5P simulations remain finite for the audited perturbations. Amplitude sweeps satisfy

$$\|y_{\text{NL}}(\epsilon) - y_{\text{LIN}}(\epsilon)\|_2 \propto \epsilon^p, \quad (54)$$

with fitted slopes 1.9820, 1.9987, and 2.0003 for representative state, governor-reference, and load perturbations, consistent with first-order linearization error.

B. NOMINAL RECONSTRUCTION AND CALIBRATION

The nominal dataset contains 20 development and 100 test trajectories, each sampled at 30 Hz for 3 s. Perturbations are normal-operation changes to governor references, loads, or their combination. The equilibrium, linear model, $P_0 = 10^{-2}I$, $Q = 10^{-6}I$, and $R = 10^{-6}I$ are frozen before TEST. B0, B1, B2, and all B3 lags are evaluated on the same trajectories. A separate 30-trajectory calibration split fits U1/U2 temperatures and low-rank discrepancy variants; TEST is not used for selection.

Hidden truth is used only for post-hoc metrics. The eight observed buses are excluded from every primary reconstruction score. The phasor prediction is formed as $V_0 + \Delta V$, and angle error is wrapped using $\text{atan2}(\sin \Delta\vartheta, \cos \Delta\vartheta)$. These rules correct an earlier exploratory implementation that incorrectly interpreted $\angle \Delta V$ as the absolute angle; no superseded result from that implementation is retained.

C. PHYSICAL MISMATCH CAMPAIGN

For each mismatch case, a fresh PowerDynamics plant is rebuilt from copied source tables, followed by power flow, dynamic initialization, and nonlinear simulation. The original package CSVs remain hash-identical. The seven families are network $R/X/B$ (M1), machine parameters (M2), governor parameters (M3), AVR parameters (M4), ZIP/load model (M5), operating-point redispatch (M6), and coupled mismatch (M7). Stress scale is $m \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$. These are engineering stress-test ranges, not probability priors.

All 980 main-grid cases and all 175 refinement cases passed power-flow, initialization, and time-domain acceptance. B1, B2, U0, U2, and the adequacy score are frozen from nominal data. Bootstrap intervals use seeds as independent

TABLE 4. Frozen experimental contracts. State and event campaigns retain separate simulators, splits, targets, and denominators.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families, seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied; exact source	303 decisions from 101 event parents $\times$ 3 seeds

units. Breakpoints in this paper are not inferred from the narrative report: the generated plots use the frozen per-case table directly because a prior refined breakpoint mixed aggregation definitions.

D. ORACLE, OFFSET, AND PHYSICAL-MAP ADAPTATION CONTRACTS

The oracle subset compares the frozen estimator with true-equilibrium recentering and true local relinearization on the same mutated trajectories. Oracle quantities are evaluation-only. The offset-only recentering study uses new DEV seeds 101–110 and TEST seeds 201–220, disjoint from all earlier datasets. Its window, cadence, regularization, and activation threshold are selected on DEV and frozen before TEST. No event labels, hidden state, or mismatch family enter the R2-A estimate.

The corrected regularized physical-recentering campaign is a separate M6-only contract. It uses 30 fresh DEV cases from seeds 601–610 and 60 TEST cases from seeds 611–630, with 20 TEST equilibria at each of $m \in \{0.5, 1.0, 1.5\}$. DEV selects the nuisance scale, regularization, and continuation initialization from a small fixed grid. TEST uses only the observed voltage and terminal-current phasors; hidden voltages are revealed afterward for scoring. A solution is accepted only when the nonlinear optimizer succeeds, all outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator; the frozen rejection table is empty. A six-case subset is also solved with an independent constrained optimizer, and its partial convergence status is reported rather than upgraded to full solver equivalence.

All numerical tables and empirical figures are regenerated from the hashed snapshot under `generated/frozen`. The build script asserts 100 nominal test trajectories, 980 main-grid B2 cases, 60 corrected-MAP TEST cases with complete acceptance, rank-deficient nuisance information, and the preregistered adaptation-closure range before producing manuscript assets.

E. EVENT-DIAGNOSIS CAMPAIGN AND SIGNAL EVIDENCE

The separate event campaign uses ANDES [47] and the same eight PMU buses. It contains 138 source configurations: one normal condition, 39 fault buses, 34 outage lines, ten generation connections, 18 load buses, eight missing-data PMUs, ten generation-and-missing pairs, eight bad-data PMUs, and

ten prescribed mixed-event configurations. Five replicas per configuration give 690 complete trajectories. Each physical trajectory lasts approximately 5 s, with the event scheduled at 2 s; missing or corrupted intervals last 0.7 s. Physical interventions are applied inside the simulator, whereas missing and corrupted values are applied to sampled measurements.

Three replicas per source configuration enter training, one development, and one test, producing 414/138/138 complete-trajectory splits. Consequently, the primary event benchmark evaluates new replicas at sources already represented in training. The separate asset-disjoint audit removes the physical source from all source-labeled fitting and tuning. Results from these contracts are reported side by side but never interpreted as the same generalization task.

The historical baseline is a hierarchy of extremely randomized trees. Its 136-feature version uses trailing voltage magnitude, angle, frequency, missingness, and a normalized disturbance score. The 448-feature version adds multi-scale statistics, adaptive residuals, and simultaneous contrasts among PMUs. A deterministic availability gate handles directly observed dropout. These are physics-guided features and typed decision heads; they are not a DAE-constrained learning architecture, and the ensemble’s leaf-vote scores are not treated as calibrated probabilities.

Figure 5 uses the first labeled episode of six representative families in the 89.65-min RAW0001 reference simulation. The curves are measured simulator outputs, not sketches or smoothed reconstructions. They show why no single channel suffices: a fault creates a short voltage sag, an outage redistributes current, injection changes have slower frequency/current responses, and measurement failures require availability or phase-consistency information.

VII. NOMINAL RECONSTRUCTION AND STRUCTURAL EVIDENCE

A. WHAT THE SPARSE PMUS RECOVER

Table 5 reports the frozen nominal TEST results. Remaining at the equilibrium (B0) gives 0.3402% mean hidden-voltage HVRE. Applying the same PMU evidence through the snapshot LMMSE update (B1) reduces this to 0.009904%, showing that the dominant nominal gain comes from the measurement-conditioned physical map rather than temporal recurrence. The causal Kalman recursion (B2) reaches 0.009634%. Because both estimators score the same 100

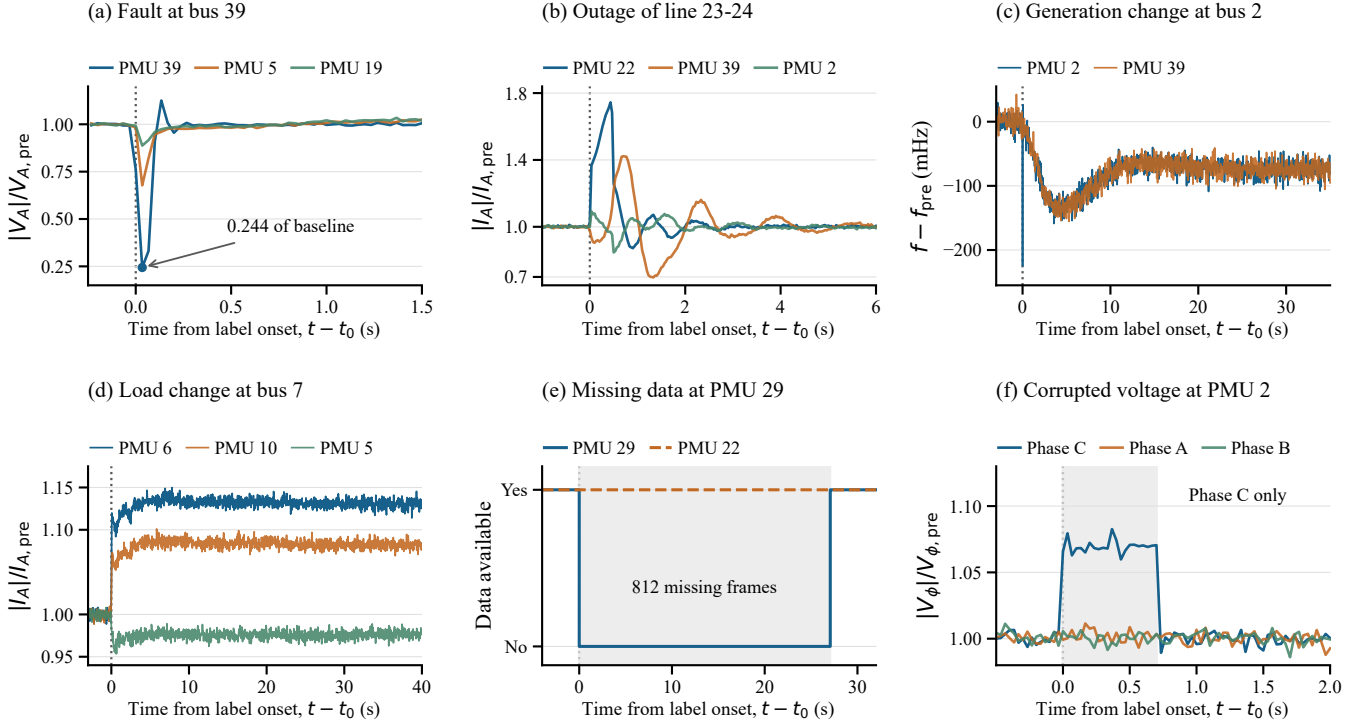


FIGURE 5. Representative measured event signatures from RAW0001. (a) Bus fault, (b) line outage, (c) generation change, (d) load change, (e) missing data, and (f) single-phase corruption. Dotted lines mark the first labeled sample; gray shading marks missing or corrupted intervals. Magnitudes are normalized by the preceding 3-s median, frequency is relative to its corresponding median, and time scales differ by panel. The plotted values and source-file hashes are frozen in the evidence snapshot.

trajectories, the relevant comparison is the paired difference: B2 minus B1 is -0.000270 percentage points (95% paired-bootstrap CI, -0.000359 to -0.000179) and B2 wins on 69.0% of trajectories. The temporal contribution is therefore consistent but small in absolute magnitude, a 2.7% reduction in the mean.

Figure 6(a) makes the trajectory distributions and scale separation visible, while panel (c) shows the paired differences rather than comparing overlapping marginal intervals. The equilibrium baseline is more than one order of magnitude worse than either measurement-updated estimator. This is not a comparison against learned or nonlinear DAE estimators; it is a controlled decomposition of equilibrium centering, instantaneous PMU conditioning, and causal temporal propagation within one local model.

B. DOES FUNCTIONAL STRUCTURE PREDICT ERROR?

At horizon 180, the finite-horizon Gramian has pseudo-rank 39 of 114 and the aggregate target residual is nonzero. Accordingly, the experiment does not declare exact full-state or exact functional observability. The preregistered bus residual nevertheless correlates with B2 HVRE at $\rho = 0.5625$ (Spearman), as shown in Fig. 6(b). A bus-level permutation test gives the two-sided p -value 0.0012, and Kendall $\tau = 0.376$. Because the 31 bus errors share trajectories and network coupling, the previously computed resampling interval [0.271, 0.751] is retained only as exploratory sensitivity, not as an

independent-unit confidence interval. The same qualitative relation appears for B1.

The relationship is informative but incomplete. Bus 36 has the largest structural residual, whereas the largest B2 errors occur at buses 33, 34, 20, 37, and 38. Thus the structural projector captures one source of difficulty but not the nonlinear excitation distribution, estimator covariance, or all numerical conditioning effects. Electrical distance to the nearest PMU has not yet been computed from the same audited PowerDynamics network, so the present analysis does not establish that $r_{F,i}$ adds information beyond that confounder. This counterexample and unresolved control are why the residual is reported as a diagnostic rather than converted into a binary observability certificate.

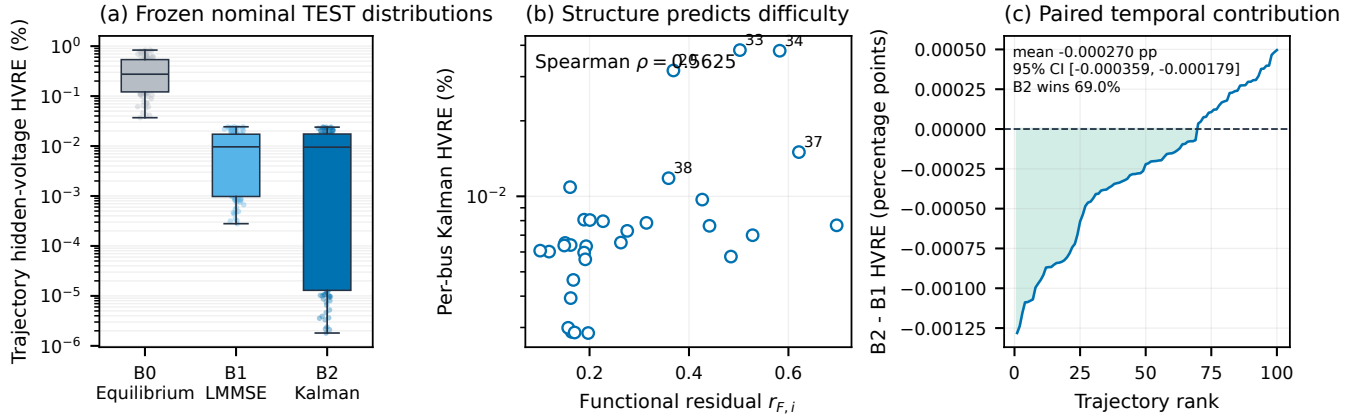
C. DOES RETROSPECTIVE SMOOTHING HELP?

Across lags from one frame to 2 s, the exact fixed-lag smoother never improves mean HVRE over lag zero; every positive lag is slightly worse in the frozen results. Figure 7(a) shows the full curve. In this regime, current voltage and terminal-current phasors already dominate the hidden-voltage update, and additional future samples do not compensate for the local-model approximation. The negative result rules out latency as a useful trade for accuracy under the tested nominal perturbations.

VIII. UNCERTAINTY, PHYSICAL MISMATCH, AND ADAPTATION

TABLE 5. Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.

Method	Complex RMSE	$ V $ RMSE	Angle RMSE ($^\circ$)	Mean HVRE (%)	95% HVRE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot LMMSE	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

**FIGURE 6.** Nominal reconstruction and structural evidence. (a) Trajectory-level hidden-voltage HVRE distributions on the same 100 nonlinear TEST trajectories. (b) Per-bus causal Kalman HVRE versus the horizon-180 functional residual exported before TEST scoring; labels identify the five buses with largest empirical error. (c) Sorted paired B2-minus-B1 HVRE differences; negative values favor the causal estimator, and the annotation reports the paired bootstrap interval and win rate.

A. MARGINAL CALIBRATION DOES NOT WHITEN INNOVATIONS

The nominal B2 covariance is highly conservative: U0 covers essentially all real and imaginary errors at the nominal 95% level. Fitting two temperatures on 30 independent CAL trajectories improves TEST hidden NLL from -13.477 to -16.184 and gives 93.9% and 94.2% marginal 95% coverage for real and imaginary components, respectively. This is useful marginal calibration, but it leaves the innovation lag-one autocorrelation unchanged at 0.602 and the Ljung–Box test rejects whiteness. Figure 7(b)–(c) and Table 6 therefore separate interval coverage from temporal model adequacy.

The calibration residual is strongly low rank: its first two principal components explain 95.7% of variance and four explain 99.2%. That empirical fact does not guarantee a useful generative correction. The rank-four measurement Gauss–Markov model reduces lag-one autocorrelation from 0.602 to 0.540, but worsens HVRE, produces hidden NLL 14.991, and undercovers both components. The rank-one process placement leaves both HVRE and autocorrelation effectively unchanged. These ablations reject the claim that low-rank colored PMU residuals, inserted without stronger physical structure, improve hidden-state uncertainty.

B. WHICH PHYSICAL MISMATCHES MATTER?

The 980-case rebuilt-plant campaign reveals sharp family dependence (Fig. 8 and Table 7). At $m = 1.5$, network mismatch reaches $16.99\times$ the pooled nominal median and operating-point mismatch reaches $8.32\times$. The coupled family

is smaller at $1.602\times$, while machine, governor, AVR, and load-model families remain near the nominal median for the tested perturbations. A blanket claim of robustness would therefore be misleading: the estimator is insensitive to several tested component shifts but fails materially when the algebraic network map or equilibrium center changes.

Nominal covariance repair does not transfer automatically. Averaged across all mismatch cases, U0 has 92.9% 95% coverage and nominally fitted U2 has only 47.5%. The reduction reflects physical shift, not a calibration/test leak: U2 remains frozen from nominal CAL data. This result limits the interpretation of Section V: nominal dispersion scaling is not robustness to a changed plant.

C. ADEQUACY SIGNALS MISMATCH AND PHYSICAL STATIC MAP RECOVERS M6

The observed-PMU innovation statistic detects many cases in which hidden HVRE is unacceptable. Macro AUROC/AUPRC are 0.901/0.779 at $m = 0.5$ and 0.928/0.787 at $m = 1.5$ [Fig. 8(c)]. Undefined family-level AUROCs, caused when every frame has the same frozen adequacy label, are omitted from the macro mean; AUPRC remains reported. Consequently, the statistic supports model-adequacy triage, not mismatch-family classification or guaranteed alarm calibration.

The earlier R2-A offset-only ablation remains scientifically useful because it isolates what does not solve the problem. Across nonnominal identifiable M1, M6, and M7 cases it closes only 0.4% of the oracle gap, despite passing its

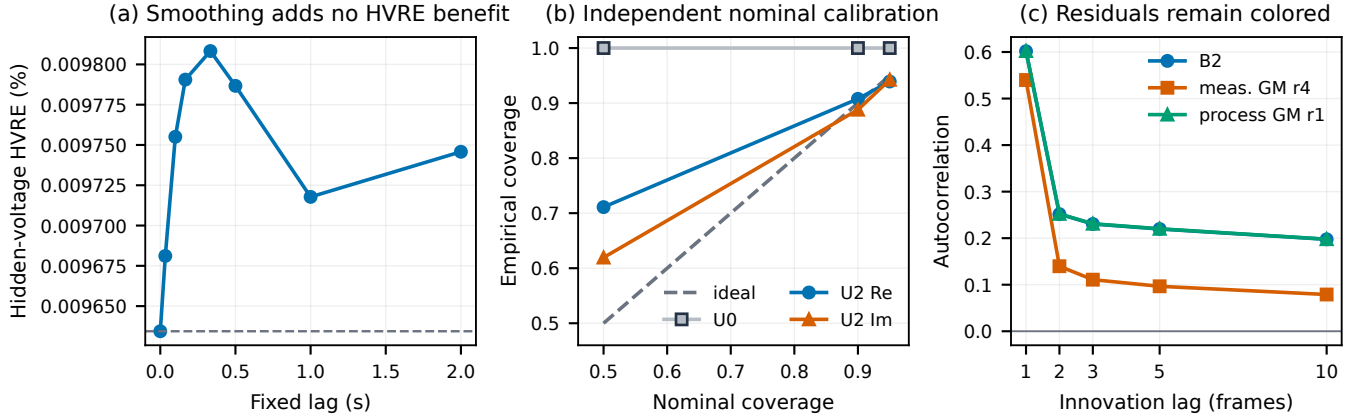


FIGURE 7. Temporal and uncertainty diagnostics on the frozen nominal TEST set. (a) Exact fixed-lag smoothing; lag zero is the causal Kalman result. (b) Marginal real-component coverage for U0 and real/imaginary U2 blocks against nominal coverage. (c) innovation autocorrelation for the original estimator and calibration-selected low-rank discrepancy placements.

TABLE 6. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

TABLE 7. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

nominal-safety gate. That failure does not imply that adaptation from observed PMUs is impossible; it shows that grouped voltage offsets without AC balance and a rebuilt physical center are inadequate.

The corrected regularized-recentering experiment directly tests that distinction for M6. All 60 TEST solves satisfy the frozen convergence gate. From $m = 0.5$ to 1.5, the nominal-center median HVRE rises from 0.19885% to 0.57795%, whereas the physical solve gives 0.00525% to 0.00974% [Fig. 9(a)]. Paired oracle-gap closure is 97.4%, 98.1%, and 98.3% across the three scales, with bootstrap intervals reported in Table 8. Median solve time is 61.7 ms and the 95th percentile is 76.0 ms. These timings characterize the implemented static solve only; they do not establish end-to-end real-time operation.

The nuisance information matrix has median rank 25 for 43 coordinates, leaving an 18-dimensional nullspace. Nevertheless, the requested hidden-voltage field is recovered with two

orders of magnitude less error than the frozen center. This is empirical evidence for target recovery despite nuisance non-uniqueness, not a proof of exact functional identifiability: the local hidden-voltage residual is nonzero at 0.0419. The distinction links the adaptation result to Section IV without claiming that the injection vector itself is recovered.

Point accuracy also does not certify posterior uncertainty. The nullspace-aware Laplace covariance attains 100% marginal 95% coverage, yet its median NEES-like statistic increases from 111.6 to 294.2 [Fig. 9(c)]. The independent constrained solver also converges on only part of its six-case subset. The E06-H conclusion is therefore strong for corrected M6 point reconstruction, partial for local target structure and uncertainty, and not established for M1 network mismatch, M7 coupled mismatch, or sequential online adaptation.

D. HOW FAR DOES THE KNOWN-SOURCE EVENT BASELINE GO?

Table 9 reports the audited sequential event baseline. On the known-source TEST split, the 448-feature hierarchy with the availability gate attains detection F1 0.989, event macro-F1 0.905, physical Top-1 0.592, physical Top-3 0.719, and integrity Top-1 0.974. The expanded representation improves physical Top-1 over the 136-feature hierarchy by 9.6 percentage points (95% paired interval, 6.9–12.5). This isolates the value of multiscale dynamics, adaptive residuals, and PMU contrasts under known-source training; the availability rule mainly improves integrity handling.

Figure 10(a) separates detection, family classification,

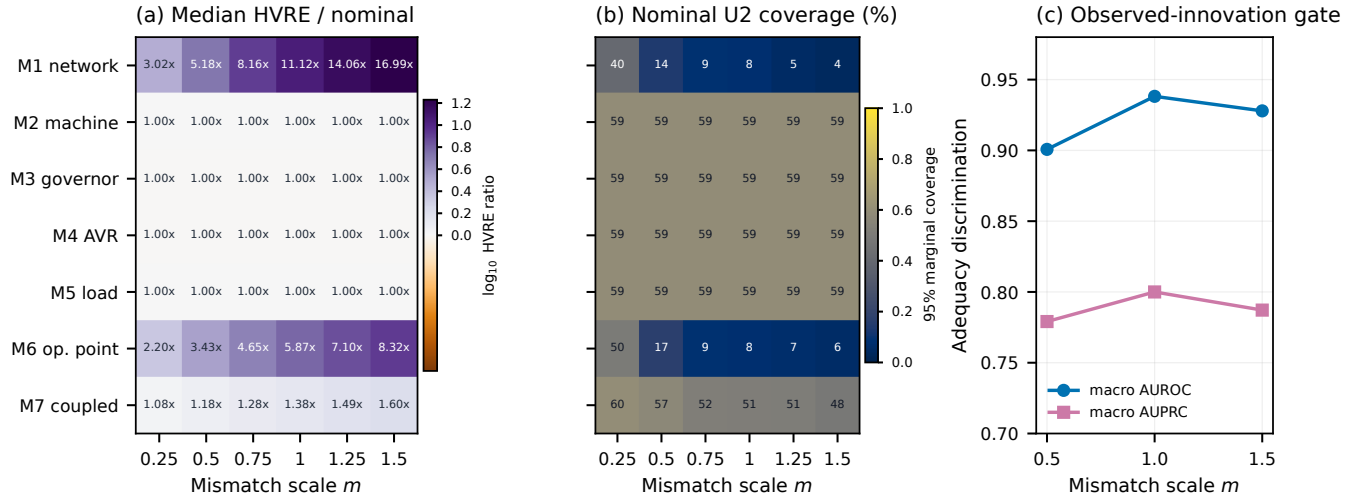


FIGURE 8. Physical-mismatch atlas for the frozen estimator. (a) Median B2 hidden-voltage HVRE relative to the pooled $m = 0$ median for every family-scale cell; annotations show the untransformed ratio and color encodes its base-10 logarithm. (b) Mean nominal-U2 marginal 95% coverage under the same rebuilt plants. (c) Macro adequacy discrimination from observed innovations; undefined family-level AUROCs are omitted from the macro mean. Each family-scale cell contains 20 independently rebuilt plants.

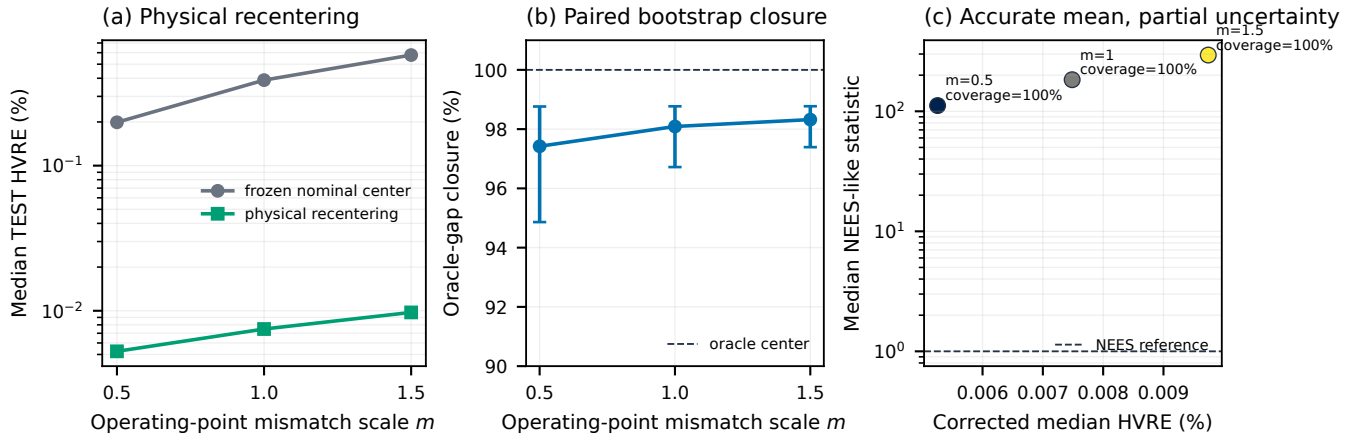


FIGURE 9. Corrected regularized physical recentering for held-out M6 operating-point mismatch. (a) Median hidden-voltage HVRE for the frozen nominal center and the physical solve. (b) Paired oracle-gap closure with 95% bootstrap intervals over 20 independent TEST cases per scale; the oracle-center reference is 100%. (c) Corrected point error versus the median joint NEES-like diagnostic. Marginal 95% coverage is 100% at each scale, but the joint statistic remains far above its nominal reference.

physical localization, and integrity localization. The gap between detection F1 and physical Top-1 confirms that a recognizable disturbance need not have a uniquely recoverable source. False alarms decrease to 1.45 episodes per normal-labeled minute, but the normal exposure is only about 7.15 min per split. That rate is evidence of alarm burden in a short synthetic stream, not an operational false-alarm guarantee.

E. DOES LOCALIZATION TRANSFER TO UNSEEN PHYSICAL SOURCES?

No. With the physical source excluded from source-labeled fitting and the true event family supplied, only 3 of 303 decisions across 101 held-source cases and three model seeds are exact, a rate of 1.0%. This is not a detector failure or a confusion among event families: it is direct evidence that

the learned location heads and candidate descriptors retain source-specific associations. The negative result invalidates any zero-shot-source claim for the legacy system.

Transfer to RAW0001 is also weak. The basic candidate model classifies an average of 2.7 of 12 event episodes and localizes 0.3 of five physical episodes across three seeds. The simulator bank omits some channels used by the longer-window implementation, so the transfer deficit cannot be assigned to one feature family alone. It nevertheless shows that improving the internal simulation score does not establish cross-record reliability.

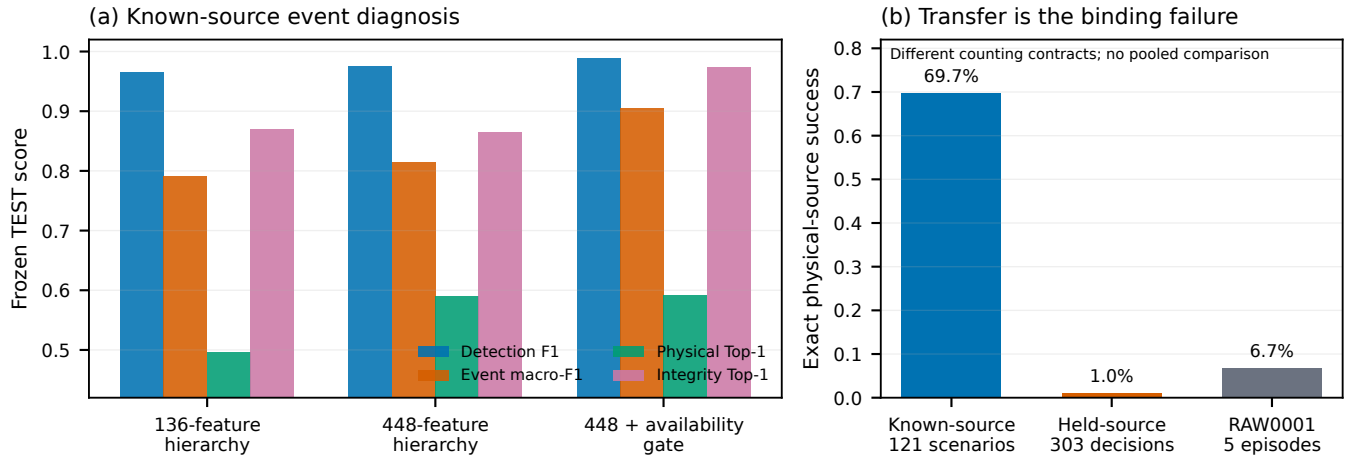
The event evidence therefore answers a precise baseline question: supervised, physics-guided feature engineering is effective for detection and known-source diagnosis, but it does not solve source-generalizable physical inference. The result motivates the candidate-conditioned residual objective

TABLE 8. Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

m	n	Frozen HVRE (%)	Corrected HVRE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

TABLE 9. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

**FIGURE 10.** Event evidence and its transfer boundary. (a) Frozen known-source TEST metrics for the compact hierarchy, expanded hierarchy, and availability-gated model. (b) Exact physical-source success under three distinct counting contracts: known-source completed scenarios, source-held-out decisions, and RAW0001 physical episodes. The bars in (b) are deliberately not pooled; they expose the change in task difficulty and data semantics.

in (35) and the profiled separation test in (28); it is not itself evidence that those proposed components work.

IX. DISCUSSION AND LIMITATIONS

The two campaigns establish different parts of the joint problem. The PowerDynamics experiments show that eight voltage/current PMUs can reconstruct the requested 31-bus voltage field accurately under the tested nominal perturbations even though the 114-state local model is far from full rank. The ANDES experiments show that the same sensor buses support strong abnormal-event detection and moderate known-source localization across eight event labels. Neither result implies the other: low hidden-voltage HVRE does not guarantee source separation, and high detection F1 does not imply a correct hidden state or physical source.

Functional observability and source diagnosability explain the distinction. Proposition 1 removes state directions that do not affect the requested voltage function. Proposition 2 instead asks whether two candidate–nuisance manifolds intersect. These are local linear results. They do not certify a nonlinear DAE over large faults, changing topology, or

unmodeled controllers. Their scientific value depends on a prospective empirical bridge: the functional residual should predict reconstruction error, and the profiled separation margin should predict candidate-set size, calibrated uncertainty, or time to source resolution.

The first bridge has partial evidence. The preregistered functional residual correlates with bus-level error, but the exceptions show that excitation, numerical tolerance, estimator covariance, and electrical distance still matter. The second bridge remains untested. The source-held-out collapse is consistent with poor physical separation or source memorization, but the present artifacts cannot distinguish those explanations. Computing (28) before training and comparing it with held-source errors is therefore more informative than adding another classifier layer.

The state-adaptation result narrows the role of identifiability. On M6 operating-point shift, regularized physical recentering closes 98.1% of the oracle-center gap although only 25 of 43 nuisance directions are locally informed. Accurate recovery of a target function can coexist with non-unique nuisance coordinates. This does not extend automatically to line

outages, faults, or sequential event-conditioned estimation. The successful solver uses one equilibrium snapshot, and its local curvature covariance remains severely under-dispersed in the joint NEES-like diagnostic.

The legacy event baseline identifies both reusable mechanisms and a hard failure. Multiscale temporal summaries, adaptive residuals, and simultaneous PMU contrasts improve known-source physical ranking by 9.6 percentage points. Availability should remain an explicit integrity input because a missing measurement is observed directly. By contrast, source identities and endpoint descriptors permit memorization: with complete physical-source holdout, exact localization falls to 1.0%. A stronger hybrid should therefore compute engineered features on candidate physical residuals, $y - \hat{y}_h$, and compare them with a physics-only likelihood. This reuses domain knowledge without treating training assets as the candidate universe.

Five limitations block a submission-ready joint-method claim. First, the state and event evidence use different simulators and are not an end-to-end common-contract comparison. Second, the PowerDynamics plant and RAW0001 are simulated; no field or hardware-in-the-loop validation is included. Third, the event bank has short, fixed-onset trajectories and about 7.15 min of normal-labeled exposure per split, which is insufficient for operational false-alarm claims. Fourth, PMU loss/join has not been tied prospectively to the information measure, and topology/source transfer remains unvalidated. Fifth, the proposed posterior in (37) has not been calibrated; classifier votes and uncorrected evidence scores must not be called probabilities.

The next frozen campaign should use one audited plant and measurement operator across all four baselines: learned-only, physics-only, physics plus learned residual discrepancy, and diagnosability-aware inference with abstention. Complete event parents and physical sources must remain disjoint across training, development, calibration, and test. The protocol should vary operating point, parameter mismatch, PMU loss/join, and at least one simulator; expose a long normal stream; report false alarms per hour, event recall, joint event-plus-source correctness, candidate-set coverage and size, source-resolution delay, HVRE, NLL/ECE, runtime, and failed solves. Until that campaign is complete, this manuscript is a consolidated scientific draft with validated components and an explicit falsifiable gap, not a completed demonstration of joint superiority.

X. CONCLUSION

This paper formulated sparse-PMU monitoring as inference of a hidden voltage field, physical event, physical source, and measurement-integrity state. Functional observability characterizes recoverability of the requested state function; affine-manifold separation gives a local diagnosability limit after state and integrity nuisance are profiled.

The consolidated evidence identifies both capability and boundary. On 100 frozen PowerDynamics trajectories, the causal estimator attains 0.009634% mean hidden-voltage

HVRE, and regularized physical recentering closes 97.4%–98.3% of the operating-point error gap on 60 held-out plants despite nuisance rank 25/43. On 690 separate ANDES trajectories, the known-source event baseline reaches detection F1 0.989, event macro-F1 0.905, and physical Top-1 0.592. Yet complete source holdout yields only 3/303 exact decisions, and transfer to RAW0001 is weak.

The central consequence is not a claim of finished joint superiority. Sparse PMUs can support accurate target-state reconstruction and strong known-source detection while still failing to identify an unseen physical source. Source-generalizable diagnosis requires physical candidate responses, an explicit integrity model, and diagnosability-aware abstention. A common-contract end-to-end campaign must establish that proposed capability.

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